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EXAMPLE

Automatic SYNCUP of S7-1500R/H controllers

S7-1500R/H / V1.0 / RH_CTRL / SYNCUP

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1. Introduction

1.1. Overview

The starting point for this application example is the RUN-Solo system state of a redundant system. The primary CPU of the system is thus in the operating state RUN and the backup CPU is in the operating state STOP.In this state, the program is only processed by the CPU in RUN.

This initial situation also occurs when you initiate the loading process of your STEP 7 project from the TIA Portal. Here the H-System switches to the system state STOP. This means that both the primary and backup CPUs are in the STOP state. This includes pausing the execution of the user program.

STEP 7 TIA Portal then loads the project or the changes made into the primary CPU in STOP and offers the TIA Portal user to start the system again. At startup, however, only the primary CPU goes back into RUN and thus the system from STOP to RUN solo - the backup CPU remains in STOP.

This is where the building block contained in the application example comes into play. This detects the RUN solo state of the system and triggers the Backup PLC to a STOP/RUN transition and thus the system to the SYNCUP.

Until now, the SYNCUP process in an S7-1500R/H system did not start automatically, but always had to be triggered manually.

1.2. Functionality

In the FB AutomaticSYNCUP, two statements are called. One is the "GET_DIAG" instruction, which can be used to read the diagnostic information of hardware components. In the case of this application example, the operating states of the two redundant controllers, as well as the system formed with them, are read out.
The second instruction is "RH_CTRL", which you can use to act on R/H systems. For example, you can lock and release the SYNCUP, or you can request it specifically. You can also request that one of the two controllers go to the "STOP" operating state. In the case of the present application example, "RH_CTRL" is used to query whether the SYNCUP is deliberately locked at the moment. If this is not the case, the SYNCUP can be requested via the user program with the same statement.

The SCL source to be found with the application example offers you the possibility that you do not have to control the SYNCUP between the two redundant controllers yourself. The code automatically checks the operating states of the primary CPU, the backup CPU, and the resulting redundant system.
This is queried with the statement "GET_DIAG". So if an error occurs and the state occurs that the system is now controlled by only one CPU, i.e. one of the two controllers is in RUN, the other in STOP and the system is thus in the RUN-Solo operating state, then the program code contained in the application example automatically recognizes this and executes the SYNCUP via the "RH_CTRL" instruction with the MODE = 7.

Before the SYNCUP is executed, the program checks whether a SYNCUP between the primary PLC and the backup PLC is locked or released (for example, via the RH_CTRL statement). If the SYNCUP is locked, it cannot be executed until the user releases the SYNCUP again via the RH_CTRL.

1.3. Components Used

This application example was created with these hardware and software components:

Component	Number	Article	Hint
S7-1518HF-4 PN	1	6ES7518-4JP00-0AB0	
PN/PN Koppler	1	6ES7158-3AD10-0XA0	
SCALANCE XF204-2BA	1	6GK5204-2AA00-2GF2	
IM 155-6 PN HF	3	6ES7155-6AU00-0CN0	Each with different IO modules
STEP 7 TIA-Portal	1	6ES7822-1A.08-	STEP 7 Professional TIA Portal V18 Upd. 1

In the application example, the automatic SYNCUP was illustrated using an H-system consisting of two S7-1518HF-4 PN controllers. However, the FB AutomaticSYNCUP serves its purpose in any R, H and HF system, regardless of which of the following four controllers this system is built with:

- CPU 1513R-1 PN
- CPU 1515R-2 PN
- CPU 1517H-3 PN
- CPU 1518HF-4 PN

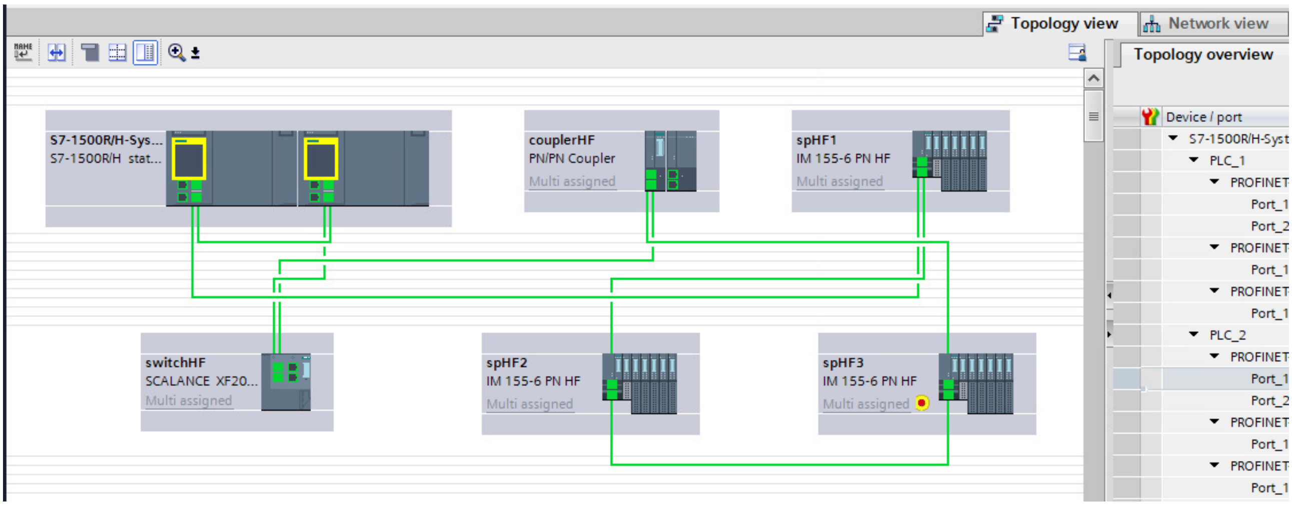
All of the above components can be purchased from the Siemens Industry Mall, for example.

This use case consists of the following components:

Component	File name	Hint
SCL-Quelle	AutomaticSYNCUP.scl	To generate a FB
UDT-Quelle	typeDiagnostics.udt	PLC Data Type for Error Handling
Documentation	Automatic SYNCUP of the S7 1500-R/H controllers	

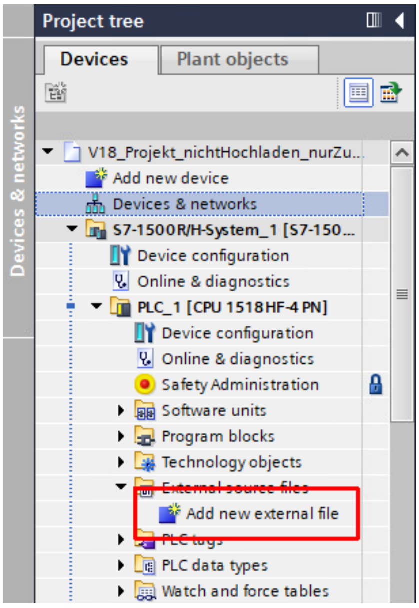
2. Engineering

2.1. Project planning and integration into the user project



Create the device configuration according to your setup. Also make sure that your PROFINET devices are not assigned to just one CPU, but have a multiple assignment, as can be seen in the example setup above. To do this, connect the PROFINET device to the PROFINET interface of both CPUs.

The UDT and SCL sources contained in the application example are imported into the TIA Portal as "External Source".

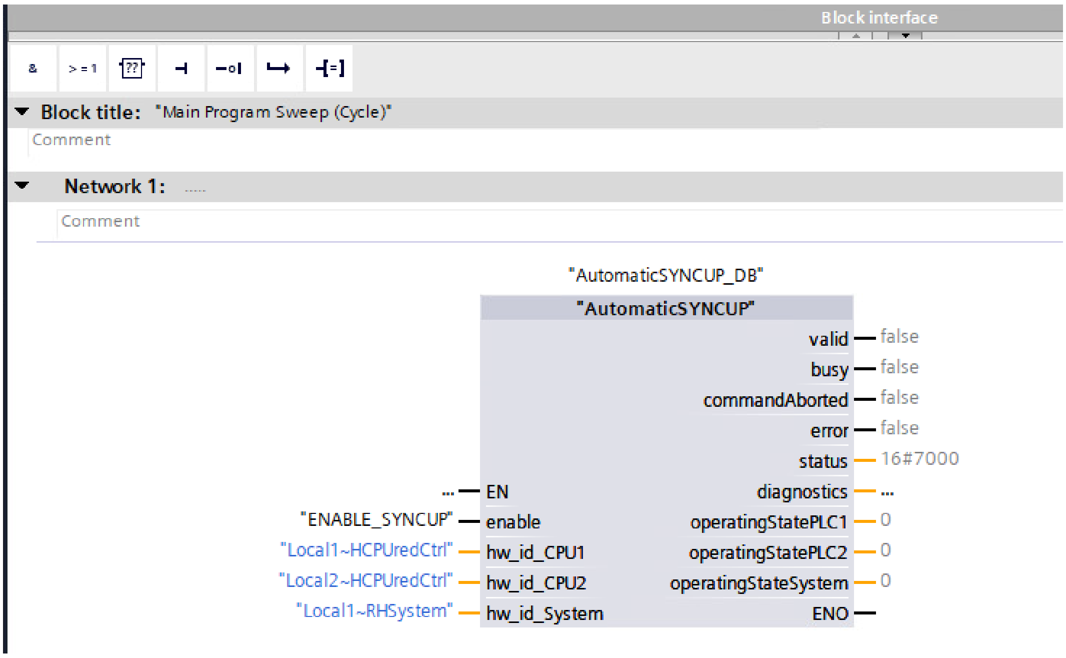


Now right-click on the SCL source and select "Generate module from source". The TIA Portal now automatically generates a function block (FB) with the name "AutomaticSYNCUP". If you call this FB in a cyclic organizational block (OB), an instance data block is also created for the FB.

If you right-click on the source "typeDiagnostics.udt" and also select "Generate block from source", a PLC data type with the name "typeDiagnostics" will be created. Translating the project into which the sources were imported should now be possible (again) without errors.

If the FB AutomaticSYNCUP is integrated into an OB and called with "enable" = TRUE, there is no longer any need to operate this FB. The building block is executed cyclically and thus checks whether there is a need for action for the building block or not. The module itself checks whether the SYNCUP can be carried out and executes it without the need for any action on the part of the user. The prerequisite is a firmware V3.0 or higher of your controllers, as only with this is it possible to query whether the SYNCUP is locked or not.

2.2. Interface description



Input	Data type	Meaning
enable	Bool	Release for the FB
hw_id_CPU1	HW_ANY	Local1~HCPUredCtrl
hw_id_CPU2	HW_ANY	Local2~HCPUredCtrl
hw_id_System	HW_ANY	Local1~RHSystem

Output	Data type	Meaning
valid	Bool	Outputs are valid/invalid
busy	Bool	Indicates whether the FB is being edited or not
commandAborted	Bool	Indicates whether the job has been canceled or not
error	Bool	Indicates whether the processing of the FB has ended with an error or not
status	Word	Returns the status code of the error that occurred
diagnostics	“typeDiagnostics”	PLC data type consisting of "status", "subfunctionStatus" and "stateNumber"
operatingStatePLC1	UInt	Operating state of CPU1 (from "GET_DIAG")
operatingStatePLC2	UInt	Operating state of CPU2 (from "GET_DIAG")
operatingStateSystem	UInt	Operating status of the redundant system (from "GET_DIAG")

2.3. Error handling

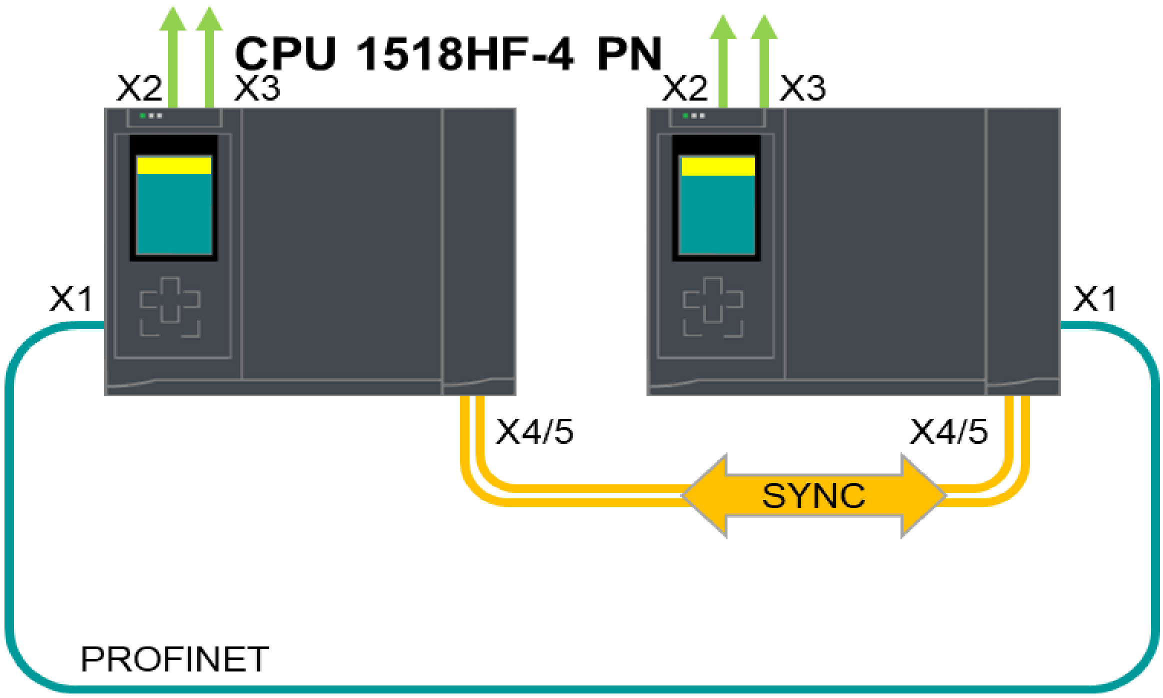
The FB "AutomaticSYNCUP" generated by the SCL source has, among other things, the two outputs "error" and "status". If the FB detects an error, the "error" bit is set to TRUE and at this point the corresponding error code is output at the output "status".

The following table lists all status codes and their meanings that the FB "AutomaticSYNCUP" can output in the event of an error:

Hexadezimalcode (16#..)	Name	Meaning
7000	STATUS_NO_CALL	FB is not being accessed at the moment.
7001	STATUS_FIRST_CALL	FB is just being accessed for the first time.
7002	STATUS_SUBSEQUENT_CALL	Follow-up call of the FB
7005	STATUS_RH_SYNCUP_ENABLED	The SYNCUP is released and not locked.
7FFF	STATUS_COMMAND_ABORTED	Request was canceled due to a new request.
8002	STATUS_ERR_INTERN_FUNCTION	Error evaluating the operating status of the CPUs.
8003	STATUS_RH_SYNCUP_BLOCKED	SYNCUP is not possible because it has been locked by the user.
8600	ERR_UNDEFINED_STATE	Undefined state reached in the step chain

3. General information about the R/H system

Basic structure of an S7-1500 system



An S7-1500R system uses the Media Redundancy Protocol (MRP) via a PROFINET ring for the redundancy of the system.

The S7-1500H system has a dedicated, redundant fiberoptic synchronisation channel. Any PROFINET topology can be used.

Both systems use two CPUs that process the same project and program data in parallel, and therefore the systems can tolerate the failure of one CPU, as the other CPU takes over control of the process.

The SYNCUP between the primary CPU and the backup CPU is needed to ensure that the two CPUs work redundantly. This happens so that in the event of a failure of the primary CPU, the backup CPU can seamlessly take over without any data loss or interruption of operations. Overall, the SYNCUP serves to ensure the redundancy and reliability of the system.

For a SYNCUP between the primary CPU and the backup CPU, certain basic criteria must be met. The prerequisites for a possible execution of the SYNCUP are listed below:

- Both CPUs have the same article number and the same firmware version.
- Each CPU contains a SIMATIC Memory Card.
- The PROFINET ring is closed and configured.
- In the redundant S7-1500H system, there is at least one redundancy connection (fiber optic cable).
- Media redundancy role in the projected PROFINET ring of an S7-1500R:

  - The two CPUs have the "Manager (auto)" media redundancy role.

  - All other devices in the PROFINET ring have the "Client" media redundancy role.
- The primary CPU is in the RUN operating state.
- The execution of the SYNCUP is not blocked (default).
- There are no charging functions running.

The SYNCUP process between the primary CPU and the backup CPU in a redundant system involves several phases that ensure that both CPUs are running the same user program synchronously again:

1. Copying the SIMATIC Memory Card: Parts of the charging memory of the primary CPU are copied to the backup CPU.
2. Backup CPU restart: The backup CPU will restart and automatically enter the SYNCUP operating state.
3. Completion of tasks: Asynchronous instructions on the primary CPU are terminated and new ones are accepted but not started.
4. Copy memory: The primary CPU takes a consistent snapshot of its memory contents and copies it to the backup CPU.
5. Catching up with the backup CPU: In this phase, the backup CPU catches up with the primary CPU.

It is important to note that too much load during SYNCUP can lead to an extension of the program cycle. Therefore, it is generally advisable to configure a sufficiently long maximum cycle time for the CPUs in an R/H system.

WARNING

- Synchronizing the data, especially copying the memory contents, increases the cycle time. This can delay the execution of the user program on the primary CPU.
- All communication links are initially interrupted during the SYNCUP. The primary CPU restores the connections after copying the memory contents, but this takes some time.
- If an error occurs in the primary CPU during the SYNCUP, it is not possible to switch to the backup CPU.

4. Appendix

4.1. Service and Support

SiePortal

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- Products & Services
  Under Products & Services you will find all our offers that were previously available in the mall catalogue.
- Support
  In the Support section, you will find all the information you need to solve technical problems with our products.
- mySieportal
  mySiePortal is your personal area that displays functions such as shopping cart management or the order overview. You will only see the full range of functions here after logging in.

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sieportal.siemens.com

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The app is available for iOS and Android:

4.2. Application Support

Siemens AGDigital Factory DivisionFactory AutomationDF FA S SUP SPH
Gleiwitzerstr. 555
90475 Nürnberg, Germany

4.3. Links and Literature

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to the contribution page of the application example https://support.industry.siemens.com/cs/ww/en/view/109975117
\3\	Link to the system manual for R/H systems https://support.industry.siemens.com/cs/ww/en/view/109754833

4.4. Change documentation

Version	Date	Alteration
V1.0	09/2024	First edition

